

EBN111 EKSAMEN/EXAMINATION (21 JUNE 2016)

ANTWOORDBLAD / ANSWER SHEET

Surname / Van:	<u>Memo</u>	Student #:							
Initials/ Voorletters:									

Q1	33	Q2	30	Q3	30	Q4		Tot:	93 /90
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QUESTION 1 / VRAAG 1 (33)

1.1	a) <u>25</u> cents ✓	b) $Q = -5 \mu C$ ✓
1.2	$P_1 = 4 mW$ ✓	$P_2 = -1.92 mW$ ✓
1.3	a) $v_{x(R=\infty)} = 27.27$ ✓	b) $v_{x(R=0)} = 30V$ ✓ c) $R_{ab} = 27 \Omega$ ✓
1.4	$R_{ab} \text{ (Circuit 1)} = 22 \Omega$ ✓	$R_{ab} \text{ (Circuit 2)} = 75$ ✓
1.5	$a = 12$ ✓	$b = 8$ ✓
	$c = 5$ ✓	$d = -9$ ✓
1.6	$e = -2$ ✓	$f = 0$ ✓
	$g = 1$ ✓	$h = -1$ ✓
1.7	a) $V_S = 6.7 V$ ✓	b) $V_{CE} = 5.53$ ✓
1.8	$\begin{bmatrix} 50 & -40 & 0 \\ -40 & 140 & 0 \\ 0 & 0 & 90 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 7 \\ 8 \\ -15 \end{bmatrix}$	

QUESTION 2 / VRAAG 2 (30)

2.1	a) $I_{x(12V)} = 1 A$ ✓	b) $I_{x(6A)} = -1 A$ ✓
2.2	$V_x = -6V$ ✓	
2.3	a) $V_{TH} = 8V$ ✓	b) $R_{TH} = 2k\Omega$ ✓
2.4	a) $I_N = -1.25 mA$ ✓	b) $R_N = 12k\Omega$ ✓
2.5	a) $P_{max} = 453.75 W$ ✓	b) $P = 8,217 W$ ✓
2.6	a) $i_x = 3.75 \mu A$ ✓	b) $R_1 = 50.05$ ✓
2.7	$V_0 = -12V$ ✓	$V_S = 8V$ ✓
2.8	$v_1 = 4mV$ ✓	$v_2 = -0.9mV$ ✓
	$v_3 = 1.67mV$ ✓	$v_o = -0.25V$ ✓

QUESTION 3 / VRAAG 3 (30)

3.1	$w_L = 25 \text{ mJ} //$	$w_C = 11 \text{ mJ} //$
3.2	$i_T(t) = 1 - 5e^{-t} \text{ A} //$	
3.3	$C_{eq} = 25 \mu\text{F} //$	
3.4	a) i_2 leads $\bar{V}_1$ by 85° degrees. //	
	b) $v(t) = 3 \cos(\omega t + 90^\circ) \text{ V} //$	
3.5	$i(t) = 0.125 \cos(4000t - 30^\circ) \text{ A} //$	
3.6	$Z_{ab} = 50 - j20 //$ (rectangular form)	
3.7	$Z_C = -8j //$	
3.8	$a_1 + jb_1 = 0 + j5 //$	$a_2 + jb_2 = 50 - j20 //$
	$a_3 = 9 //$	$a_4 = -10 //$
3.9	$\bar{V}_0 = -4.8 + 3.6j \text{ V} //$ (rectangular form)	
3.10	$\bar{I}_X = 0.8 + 1.6j \text{ A} //$ (rectangular form)	
3.11	$\bar{V}_{TH} = 4 \angle -135^\circ //$ (polar form)	
	$Z_{TH} = 2 + 2j //$ (rectangular form)	